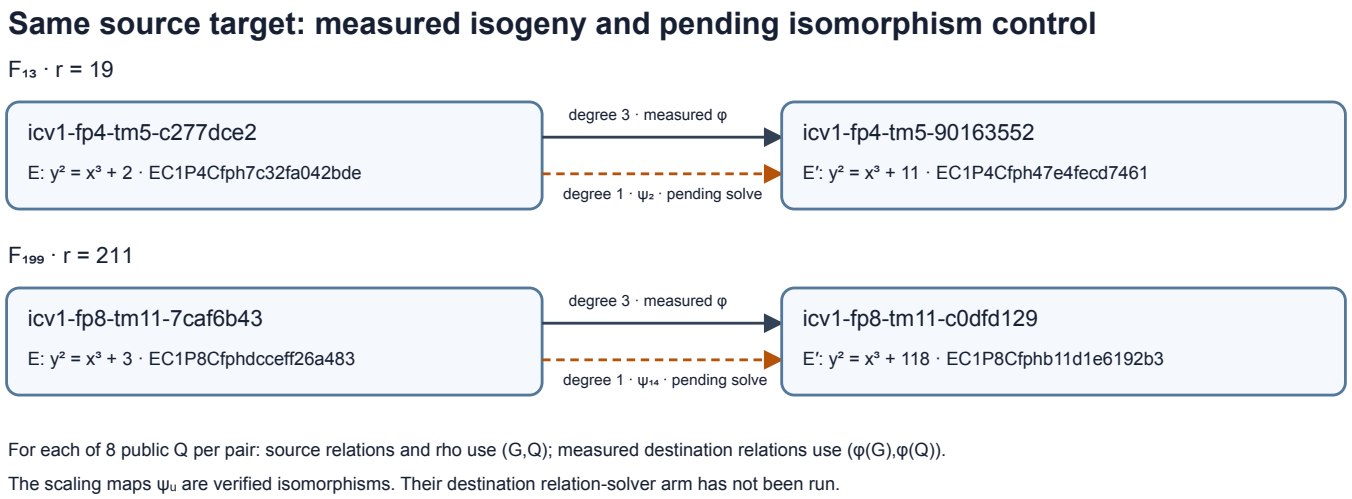
Explicit isogeny transport: same-target relation solves

2026-10-12 local · final run 2026-10-13 02:00 UTC · T-2223 / T-2240

The degree-three map on each registered prime-order pair has a geometric kernel of size three and a rational kernel of size one. We tested the same eight public source targets per pair with source relations, transported relations, and the repository's negation rho. All 48 arm answers verified. Full rational-point group-law checks passed 361 and 44,521 ordered pairs. Final structural review found public degree-one scaling isomorphisms to both destination models; an isomorphism-only solver arm is still owed.

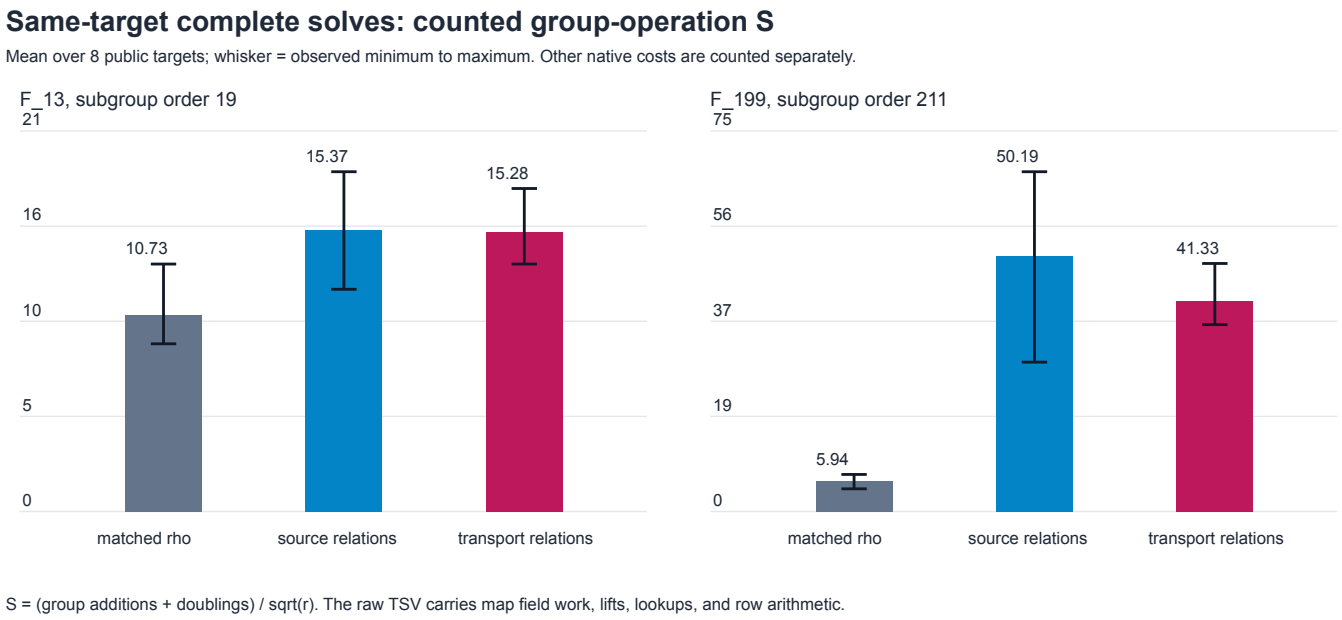


The source and destination use the exact full models and EC1 generators listed above. The map is $\varphi(x,y) = ((x^3+4b)/x^2,\ y(x^3-8b)/x^3)$. For the standard construction, see Shumow, **Isogenies of Elliptic Curves: A Computational Approach**, arXiv:0910.5370. Full model tuples and frozen inputs are in `curves.json` and `PROTOCOL.md`. The scaling $\psi_u(x,y) = (u^2x, u^3y)$ uses $u = 2$ at $p = 13$ and $u = 14$ at $p = 199$. The independent verifier checked both maps on 44,882 ordered source-point pairs. The diagram marks this verified but unmeasured arm.

Counted group-operation S is (additions + doublings)/$\sqrt{r}$. The reference is matched negation rho on the exact source Q. Its two-class walk-step floor is $\sqrt{(\pi/4)} \approx 0.88623$ S. Each curve also has a rational order-three automorphism ($\zeta = 3$ for F13, $\zeta = 92$ for F199), so the six-class curve floor is $\sqrt{(\pi/12)} \approx 0.51166$ S. Six-class rho was not run. Ranges below are observed min–max over eight targets, not confidence intervals. Ratios are sums over matched targets.

r	arm	mean S (range)	S/rho	S/A2 floor
19	negation rho	10.725 (9.177–13.536)	1.000	12.102
19	source relations	15.371 (12.159–18.583)	1.433	17.344
19	transport relations	15.285 (13.536–17.665)	1.425	17.247
211	negation rho	5.938 (4.475–7.297)	1.000	6.700
211	source relations	50.186 (29.396–66.846)	8.452	56.629
211	transport relations	41.332 (36.762–48.810)	6.961	46.638

Phase evidence and limits



At order 211, mean counted group operations per target are 729.000 for source relations, 600.375 for transport relations, and 86.250 for rho. The source/transport ratio is 5,832/4,803 = 1.21424 in this counted unit. Relation collection uses 525.25 versus 387.75 group operations; the mean trial counts are 61.50 versus 44.875. Both reach rank 12. The same destination curve is reachable through ψ_u, so this yield gap has not been isolated as a degree-three isogeny effect. Setup is 78 operations each; target descent is 23.25 versus 23.125; verification is 102.50 versus 111.50. Rho uses 77.75 operations in its walk and 8.50 in the runner check, on average.

The destination map evaluates G and Q per workload, costing 10 field multiplications, 4 squarings and 2 inversions in this implementation. Factor-base square roots, lookup work, modular row operations, and rho distinguished-point table probes lack pinned conversion into S here. The chart is therefore a counted-operation diagnostic, not an admitted complete-cost comparison. No wall-time claim is made on this shared Darwin arm64 host (L0). For the six-class curve floor, multiply the table's last column by √3. The order-211 values are 11.605, 98.085 and 80.779. The selected negation rho is an exact-target control, not a certified best rho on these curves. This correction changes no raw measurement.

Final `raw-v4.tsv` and `replay-v4.tsv` compare byte for byte, SHA-256 `995b7bcc33f5584d7b4fc6102b7ade2b3f19c7e5b39cb406b0b1c33dc7af11ad`. This is same-host replay. Earlier v1–v3 runs are retained with their phase-label or output-schema defects in `reports/`; only v4 supports this table. Source, commands, full phase counters, run reports, and requirement status are in `REPORT.md`. A separate native verifier with its own point law passed all 48 records, checked the four group orders and all phase sums, checked the cubic automorphism on 89,764 ordered point pairs, and scaling isomorphisms on 44,882 pairs; its receipt is `validation/independent_arithmetic_v4.log`. Lean checked concrete cubic roots, sixth-power scalings and conditional equation identities in `lean/CubicAutomorphism.lean`. General Lean group-action theorems remain open. The controlled two-thread, loopback-accessible release library suite passed 4,009 tests with none failed and 113 ignored. The earlier restricted-sandbox run failed 13 and remains separately recorded.

Immediate mechanism check: solve the same 16 targets after ψ_u transport with equal relation seeds, factor bases and caps; include the original arms and verify all scalars. Then add charged six-class rho. A later native paired-target ecbench measurement needs growing sizes, all cold costs, independent-host replay, and a non-isomorphic pair to isolate the map's specific effect. The gap is tracked in `problems/nonrational-isogeny-fiber-dlp-cost-gap.md`.